ROAMLY

Community-Driven Travel Discovery Platform

Project Proposal

ICT 2022 | Full Stack Development

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1. Introduction

Roamly is a community-driven travel discovery and local listing platform designed to bridge the gap between travellers seeking destination information and local businesses offering services within those destinations. Built initially for Sri Lanka, one of Asia's most diverse and rapidly growing tourism markets, Roamly is architected for global scalability.

Sri Lanka hosts millions of tourists annually who visit iconic destinations such as Galle, Ella, Sigiriya, Mirissa, and Kandy. Despite the volume of tourism, travellers frequently struggle to find reliable, consolidated information about local attractions, budget estimates, dining options, accommodation, and equipment rentals in a single place. Existing solutions are fragmented, outdated, or not locally focused.

Roamly addresses this by providing a two-sided platform: visitors can browse and contribute destination content, while local business owners can register and list their services , all moderated by an administrator to ensure quality and accuracy.

2. Problem Statement

Travellers visiting Sri Lankan destinations face several interconnected challenges:

- Information is scattered across multiple websites, social media groups, and travel blogs with no single source of truth for a given destination.
- There is no easy way to discover smaller, lesser-known tourist spots that locals consider beautiful or significant, these are often shared only through word of mouth.
- Budget planning is difficult as pricing for activities, food, accommodation, and rentals is rarely consolidated or compared in one place.
- Local businesses such as surf board rental shops, homestays, guided tour operators, and restaurants have limited digital presence and no structured platform to reach inbound tourists.
- Existing platforms like TripAdvisor are global and generic, they do not cater to the granular, hyperlocal needs of Sri Lankan tourism destinations.

Roamly solves this by creating a community-powered platform where travellers and local owners collectively build and maintain destination content, under admin oversight, making local tourism information accurate, comprehensive, and accessible.

3. Objectives

- Develop a full stack web platform that allows visitors to browse Sri Lankan destinations with consolidated information including attractions, food, accommodation, and rentals.
- Enable community-driven content creation where both visitors and local business owners can submit new tourist spots, upload photos, and write descriptions.
- Provide local business owners with a self-service portal to register, list, and manage their services, including pricing, availability, and photos.
- Implement a three-tier role system (Visitor, Business Owner, Admin) with appropriate permissions and an admin approval workflow for submitted content.

- Allow visitors to filter destinations and listings by budget range to facilitate affordable trip planning.
- Build a scalable architecture that can be extended to destinations beyond Sri Lanka in future iterations.
- Deliver a responsive, user-friendly interface accessible on both desktop and mobile browsers.

4. Proposed Features and Functionalities

4.1 Visitor Features

- Browse destinations (e.g. Galle, Ella, Mirissa, Sigiriya) with rich detail pages
- View places to visit with opening hours, entry fees, and travel tips
- See places to eat, stay, and rent equipment with pricing
- Filter all listings by budget range
- Submit a new tourist spot with photos, description, and location
- Leave reviews and star ratings on listings and tourist spots
- Save favourite destinations and spots to a personal wishlist
- Register and manage a personal profile

4.2 Business Owner Features

- Register a business profile with contact details and location
- Create and manage listings (stays, rentals, tours, restaurants) with photos and pricing
- Set and update availability and operating hours
- Submit new tourist spots relevant to their area
- View reviews received on their listings

4.3 Admin Features

- Review and approve or reject tourist spot submissions from visitors and owners
- Verify and approve business registrations and listings before they go live
- Moderate reviews and uploaded photos
- Manage destination categories and platform-wide content
- View platform analytics — total users, listings, submissions, and activity by destination

4.4 Platform-Wide Features

- Responsive design — works on desktop, tablet, and mobile browsers
- Search and filter by destination, category, and budget
- Photo gallery support for destinations and listings
- Pending approval workflow — submitted content is held for review before going public

5. Technology Stack

The following technologies have been selected based on suitability for a full stack web project, ease of local deployment, and examiner familiarity:

Layer	Technology	Purpose
Frontend	HTML5 / CSS3	Structure and styling of all web pages
	JavaScript (Vanilla)	Client-side interactivity and dynamic UI
Backend	PHP	Server-side logic, routing, and API handling
Database	MySQL	Relational data storage and querying
Server	Apache (XAMPP)	Local development and deployment environment
Version Control	Git / GitHub	Collaborative development and code management
Design	Figma	UI/UX wireframing and prototyping

PHP was chosen as the backend language due to its natural integration with HTML, its widespread use in academic full stack projects, and the ease of deploying with XAMPP. MySQL provides a structured relational model well-suited to the multi-role, multi-entity nature of this platform.

6. System Architecture Overview

Roamly follows a classic three-tier web architecture:

Tier	Description
Presentation Tier	HTML/CSS/JavaScript pages rendered in the user's browser. Separate views for Visitor, Business Owner, and Admin roles.
Application Tier	PHP processes all business logic — user authentication, role checking, content approval workflows, and data validation before querying the database.
Data Tier	MySQL database stores all users, destinations, tourist spots, business listings, reviews, photos, and approval statuses.

User Flow Overview

- Visitor registers → browses destinations → filters by budget → submits a new spot → spot enters pending queue
- Business Owner registers → admin approves account → owner creates listing → listing enters pending queue → admin approves → goes live

- Admin logs in → reviews pending spots and listings → approves or rejects → manages platform content

7. Database Overview

The MySQL database will consist of the following core tables:

Table	Key Fields & Purpose
users	user_id, name, email, password_hash, role (visitor/owner/admin), created_at
destinations	destination_id, name, region, description, cover_image, created_by_admin
tourist_spots	spot_id, destination_id, submitted_by, name, description, category, opening_hours, entry_fee, status (pending/approved/rejected), created_at
spot_photos	photo_id, spot_id, uploaded_by, file_path, uploaded_at
businesses	business_id, owner_id, name, category (stay/rental/food/tour), destination_id, description, price_range, status, created_at
listings	listing_id, business_id, title, description, price, availability, photos, status
reviews	review_id, user_id, target_id, target_type (spot/listing), rating, comment, created_at
wishlists	wishlist_id, user_id, spot_id, added_at

Relationships are enforced through foreign keys. The status field on tourist_spots and businesses drives the admin approval workflow — only approved records are visible to the public.

8. Project Timeline

The project is planned across a nine-week development cycle:

Week	Phase	Tasks
1	Planning & Design	Finalise requirements, create wireframes in Figma, design database schema
2	Environment Setup	Set up XAMPP, GitHub repo, folder structure, base HTML templates and CSS
3	Authentication	User registration, login, role-based session management (visitor/owner/admin)
4	Destinations & Spots	Destination pages, tourist spot listings, photo upload, opening hours display

5	Business Listings	Business owner registration, listing creation, pricing and availability management
6	Visitor Features	Budget filtering, spot submission form, reviews and ratings, wishlist
7	Admin Panel	Approval workflows for spots and listings, content moderation, analytics dashboard
8	Testing & Refinement	End-to-end testing across all roles, bug fixing, UI polish, performance checks
9	Documentation	Final report, user manual, deployment guide, presentation preparation

9. Expected Outcomes

Upon successful completion, Roamly will deliver:

- A fully functional full stack web application with role-based access for Visitors, Business Owners, and Administrators.
- A community-driven destination database covering multiple Sri Lankan locations, with user-submitted tourist spots and photo galleries.
- A business listing marketplace where local owners can independently manage and promote their services to inbound tourists.
- An admin content moderation system ensuring all publicly visible content is verified and accurate.
- A budget-aware discovery experience that helps travellers plan trips within their financial means.
- A scalable codebase and database schema that can be extended to international destinations, positioning Roamly for global expansion beyond its initial Sri Lanka focus.

Beyond academic outcomes, Roamly represents a genuinely viable solution to a real problem in Sri Lankan tourism — one that could be developed into a production product, supporting local economies and improving the travel experience for both domestic and international visitors.

10. References

No external references were used in the preparation of this proposal. All platform concepts, system design decisions, and feature specifications are original work developed by the project group.

The following tools and documentation were consulted during the planning phase:

- PHP Official Documentation — https://www.php.net/docs.php
- MySQL Reference Manual — https://dev.mysql.com/doc/
- MDN Web Docs (HTML, CSS, JavaScript) — https://developer.mozilla.org/
- Figma Design Tool — https://www.figma.com/
- W3Schools Web Tutorials — https://www.w3schools.com/